

### Exercise n. 4

Write a code to compute  $\pi$  based on the numerical integration of the following integral using the trapezoidal rule:

$$\pi = 4 \int_0^1 \sqrt{1-x^2} dx.$$

Recall that, given a generic integration interval  $[a, b]$  discretized using  $N$  nodes, the trapezoidal approximation of a generic function  $f(x)$  reads:

$$\int_a^b f(x) dx = \frac{b-a}{N} \sum_{k=0}^{N-1} \frac{f(x_k) + f(x_{k+1})}{2}.$$

Carry out the following tasks:

1. Write a serial version of the code in double precision.
2. Verify that the serial code returns the exact expression of  $\pi$  (up to machine precision). Recall that  $\pi = \arccos(-1)$ .
3. Enable parallel computation using the MPI library, allowing execution on an arbitrary number of processes. To do so, the integration interval must be divided into subintervals equal to the number of processes used in the simulation.
4. Verify that the result is fully independent of the number of processes used.
5. Measure the performance of the code in terms of execution time (for this, it is recommended to use a large number of points for grid discretization, e.g. 10M), varying the number of processes from 1 to  $N_{MPI}$ . In particular, perform both strong scaling tests (fixed  $N$ ) and weak scaling tests (with  $N$  proportional to  $N_{MPI}$ ), and plot both speed-up and scaled speed-up, respectively.
6. Fit the results using Amdahl's Law and Gustafson's Law to estimate the values of  $p$  and  $s$ , remembering that  $p + s = 1$ .

**Note:** The report can be done individually or in pairs. Conciseness, clarity of presentation, and quality of images are appreciated.